

MOHAMMED GHALIB

Software Engineer • Backend, Web & Mobile • Applied AI, RAG & Agentic Systems

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[Portfolio](#) • [LinkedIn](#) • [GitHub](#)

PROFESSIONAL SUMMARY

Software engineer who builds and ships complete products, not just features. I write the Django backends, the React web and React Native mobile apps on top of them, and the auth, payments, and background jobs underneath, then containerise, deploy, and keep the whole thing running in production. I'm comfortable owning a product from the first architecture decision through to a live, maintained release, working directly with the client it's for.

I build the AI too, and I build it to be reliable. When a product needs it, I add retrieval and agentic systems that plan their own lookups, ground every answer in real sources, verify their own output, and refuse to guess when they aren't sure. That's the core of Sheria AI, a Kenyan-law assistant, and Ummly, an Islamic knowledge platform, and I've taken the same approach into client work across healthcare, education, and IoT.

I've owned an AI legal product end to end at Datawise Africa, led software and AI delivery for clients at Digital Appex, and supported enterprise systems at a national bank. Across all of it I stay hands-on across the whole stack: backend, frontend, deployment, and the person the work is for.

CORE SKILLS

Languages & Frameworks: Python, TypeScript/JavaScript, Django, Django REST Framework, FastAPI, Celery, React, React Router (SSR), React Native (Expo), Tailwind CSS, Redux Toolkit, TanStack Query

AI Engineering: Agentic pipelines on LangGraph (plan, research, verify, repair), tool-calling, RAG design, LangChain, vector databases (Pinecone, ChromaDB), chunked embeddings, reranking (bge-reranker), similarity thresholds, grounding & citation, hallucination control, prompt engineering, evaluation harnesses

LLMs: OpenAI, LLaMA (via Groq), Google Gemini, DeepSeek, reasoning models (gpt-oss), multi-model routing and fallback

APIs & Integrations: REST APIs, JWT (RS256 / SimpleJWT), WebSockets/Channels, Paystack, M-Pesa STK Push, PayPal, Stripe, ZeptoMail, SMS/email gateways, KRA iTax

Data & Infra: PostgreSQL, Redis, SQLite, vector stores; Docker & docker-compose, Gunicorn, Nginx (nginx-proxy + Let's Encrypt), DigitalOcean, Vercel, Netlify, Render, Supabase

DevOps & Quality: GitHub Actions CI/CD, SonarQube quality gates, PyTest + coverage, Vitest, semantic-version releases, staging/prod environments, Git, Agile/Scrum, Jira, Slack

Enterprise: Oracle Fusion HCM/Financials, RBAC, user provisioning, security policies

PROFESSIONAL EXPERIENCE

Software Engineer, Datawise Africa

2026 • Remote

Kenyan AI and software company. I own an AI legal product end to end, and deliver client and internal AI systems alongside it.

- Own **Sheria AI** ([sheria.africa](#)), an AI legal assistant for Kenyan law, from architecture to live deployment. Rebuilt its core from a single retrieve-then-answer call into an **agentic pipeline on LangGraph**: each question is triaged, then researched by a model that plans its own lookups against case-law and Constitution tools, and each answer is drafted, verified, and repaired before the user sees it, with a coverage check so no part of a multi-part question is dropped.
- **Made retrieval genuinely work.** Chunked the case index so search reads the court's actual reasoning instead of just the header, tuned the relevance thresholds that keep irrelevant cases out of answers, and enforced grounded citations with a firm no-fabrication rule and a graceful refusal when nothing relevant is found.
- **Built the evaluation harnesses** the team ships on: a triage scorecard, and a blind side-by-side comparison of candidate answer models. Prompt and model changes are backed by evidence, not guesswork. The judgment lives in editable prompt files, so behaviour is tuned in prose rather than code.
- **Own the frontend**, built with React Router (server-side rendering), TypeScript, Redux Toolkit, and TanStack Query: a streaming chat UI that renders answers as they generate, with ranked source cards.
- **Deliver AI systems for clients and internal teams** beyond Sheria AI, and run the delivery pipeline: CI, a quality gate, test coverage, and versioned releases.

Software Engineer, Digital Appex

2025 • Remote

Kenyan technology company delivering software and AI solutions. I lead the platform engineering and client delivery.

- Built and now own [digitalappex.co.ke](#), a **Django / DRF platform** covering products, orders, quotes, payments, warranties, messaging, projects, and a chatbot, with a React and Vite frontend. It's live and I keep it there: ongoing feature work, maintenance, and releases.

- **Deliver software and AI for the company's clients**, turning customer requests into shipped systems. This includes AI-driven solutions for organisations in healthcare and education, alongside the company's own products.
- **Built the platform's foundations**: JWT auth, Paystack payments, role-based access, and background jobs on Celery and Redis. Containerised with Docker and deployed to a DigitalOcean server behind nginx-proxy with automated SSL and CI on GitHub Actions.
- **Set the engineering standards** the company builds on: architecture, documentation, and the release process.

Financial Systems Intern, Kenya Commercial Bank (KCB)

Jan 2025 – Jul 2025 • On-site, Nairobi

Production support for Oracle Fusion at one of Kenya's largest banks.

- Provided daily production support for Oracle Fusion HCM and Financials serving 1,000+ staff.
- Created and governed user roles, access restrictions, and security policies, enforcing internal controls and audit requirements (RBAC).
- Designed and exposed secure REST API endpoints integrating Oracle Fusion with internal services and external systems, including KRA iTax and payment gateways.
- Resolved tickets within SLA and coordinated fixes across HR, Finance, and IT stakeholders.

Software Engineer, Bee MultiScent Ltd

Feb 2023 – Dec 2023 • Hybrid

Scent-technology company. I ran the internal platform, the AI diffuser work, and the website.

- **Built the company portal**: a Django / DRF backend with a React frontend and real-time dashboards used across operations, HR, and inventory, with role-based access, task assignment, document management, and WebSocket notifications, and a companion React Native mobile app.
- **Developed the AI for a smart scent diffuser**, automating how and when the company's IoT diffuser released scent.
- **Built and maintained the company website**, and owned ongoing development and maintenance across the company's systems.
- Earned an official recommendation letter for performance and teamwork (available on request).

Independent Software Engineer (Contract), Private Clients

Jan 2024 – Aug 2025 • Remote

End-to-end full-stack platforms designed, built, and deployed for clients.

- **Ummly (formerly Islamify), Islamic knowledge platform** (ummlly.co.ke). Built the whole product solo across **backend, web, and mobile**, built around a **Pinecone RAG** assistant grounded in the Qur'an and Hadith: reranking, a confidence floor that returns a low-confidence message instead of guessing, intent classification, query rewriting, bilingual EN/AR answers, and per-excerpt citations. LLaMA via Groq with a Google Gemini fallback. Containerised and deployed through GitHub Actions to a DigitalOcean server, with email and Paystack payments; a React Router (SSR) web app on Vercel and an Expo / React Native mobile app.
- **Collaborative Approach, a referral & tour-booking SaaS** (collaborativeapproach.africa). React + DRF; integrated M-Pesa STK Push, PayPal, and Stripe; referral tracking with automated commission calculation, JWT auth, Celery background tasks, and role-based portals (Admin / Agent / Customer).
- **Sheikh Siraj Rahman College, a school management system** (sheikhsirajrahman.ac.ke). Multi-portal (Admin / Teacher / Student / Parent) on React, Django, and PostgreSQL, with online admissions, fee payments, results publishing, and SMS/email automation, deployed on HostPinnacle with a CI/CD pipeline.

EDUCATION

Bachelor of Business Information Technology, Mount Kenya University

2022 – 2025

Major in Information Technology, Software Development, and Artificial Intelligence.

CERTIFICATIONS

- Build AI-Powered Apps, Code with Mosh
- Ultimate Django Series, Code with Mosh
- Ultimate Docker Course, Code with Mosh
- Ultimate React + TypeScript Course, Code with Mosh

REFERENCES

Available on request.